

Bram J. Banik

brambanik@gmail.com | www.linkedin.com/in/bram-banik | 778 872 5628

Website: brambanik.github.io | Github: github.com/brambanik

Education

BASc in Engineering Physics - University of British Columbia

Expected April 2028

- Dean's List - 84.5%
- Notable Courses: Differential Equations, Circuit Analysis II, Multivariable/Vector Calculus, Signals and Systems, Electromechanical Energy Conversion, Quantum Mechanics, Electricity and Magnetism

Experience

UBC Thunderbikes

Aerodynamics Team Lead

August 2025 - Present

- Spearheading a subteam of 10+ members for first entry into the MotoStudent competition
- Leading carbon fiber fairings, rear frame, gas tank, and projects
- Managing \$4000+ budget for carbon fiber, MDF, epoxy, and machinery purchases
- Training members in Solidworks CAD modelling, Fusion360 CNC routing, and aerodynamics simulations

Aerodynamics Subteam Member

September 2024 - July 2025

- Designed and built a carbon fiber gas tank and battery panels for an all-electric motorcycle
- Operated CNC routers and waterjet cutters for mould fabrication
- Used wet-layup and vacuum infusion to manufacture composite gas tank

Engineering Physics Student Council

Student Life Coordinator

July 2025 - Present

- Organizing 8 ongoing community events for the Engineering Physics student body
- Managing a budget of \$1000+ to distribute annually
- Reigniting interest in social events, leading to improved student attendance

Projects

Fully Autonomous Pet-Rescuing Robot

- Engineered an autonomous robot using a computer vision + ESP32 stack
- Used PyTorch to train YOLO computer vision model, communicating between a Raspberry Pi and ESP32
- Designed and soldered 4 custom PCBs to handle sensors and motor control
- Wrote codebase in C++ using freeRTOS to allow efficient time-slicing
- Debugged and handled noise issues from drive motors using shielding and software polling

IL Driving Model

- Navigated obstacle-rich ROS environment in <1 minute using computer vision model
- Used Tensorflow to train multiple lightweight, task-focused CNNs to drive simulated car
- Processed 30,000+ images for training using augmentation to enhance dataset
- Created Python state-machine logic to handle obstacle detection and avoidance

Analog Motor Control Circuit

- Built and tested a potentiometer circuit to control the speed of a motor
- Tracked motor RPM using counter and D-Latch
- Wrote detailed lab reports to track bugs and circuit improvements
- Used circuit analysis techniques and oscilloscopes to debug circuit

Skills

Electrical: KiCad, PCB Design, Verilog, VHDL, Simulink, Soldering, Oscilloscopes

Mechanical: Solidworks, OnShape, ANSYS, CNC Mill, Laser and Waterjet Cutting, Carbon Fiber Wet Layup and Vacuum Infusion

Software: Git, C, C++, Java, MATLAB, Python, HTML + CSS, Linux, RTOS

ML: PyTorch, Tensorflow, OpenCV, CNNs, NumPy, CUDA C